

EDUCATION

Georgia State University

Atlanta, GA, USA

Master of Actuarial Science

August 2027[†] – December 2028 (*Expected*)

National Yang Ming Chiao Tung University

Hsinchu City, Taiwan

Bachelor of Science in Applied Mathematics

August 2022 – June 2026

École polytechnique

Palaiseau, France

International Exchange Program

September 2024 – June 2025

SKILLS

Programming Python, C++, and R; $\text{\LaTeX}$ and **typst**.

Languages Mandarin, English (CEFR C1), Japanese (JLPT N2), and French (conversational).

Actuarial Pathway SOA Exam P (July 2025), FM (December 2025), VEE Mathematical Statistics, and VEE Accounting & Finance.

COURSEWORK

Survival Analysis[‡], Time Series[‡], Machine Learning[‡], Probability Theory, Advanced Statistics, Econometrics, and Futures & Options.

PROJECTS

Motor Insurance Pure Premium Pricing: Bridging Machine Learning and Actuarial Compliance using Tree-GLM Hybrids

March 2025 – Present

Independent Research Project

Submitted to NAAJ and under review

- Developed a three-stage pipeline of rule extraction, binning, and penalized GLM to evaluate pure premiums using the `freMTPL2` dataset.
- Verified that the additive structure of Tree-GLM imposes no statistically detectable risk-ranking penalty relative to LightGBM while preserving inspectability.
- Showcased that Tree-GLM's risk-ranking is dominated by behavioral (`BonusMalus`) rather than demographic covariates (`Region`).

Study of Hitting Time and Probability for Random Walks in One and Two Dimensions[§]

July 2025 – February 2026

Research Project, College Student Research Scholarship, [National Science and Technology Council](#)

- Derived hitting time distributions and expectations for one-dimensional and two-dimensional simple random walks.
- Established an N^2 scaling for expected hitting times in bounded domains and identified the discrete probability coefficients with the centroidal value of the Prandtl stress function from the torsion problem, providing a cross-disciplinary analytical framework for evaluating boundary behaviors in random walks.

The Gambler's Ruin Problem for One-Dimensional Random Walks: Simple Symmetric Random Walk and Some Extensions[§]

September 2025 – December 2025

Semester Project, [Individual Directed Study \(I\)](#), National Yang Ming Chiao Tung University

- Derived the x/r hitting probability for the simple symmetric walk via the optional sampling theorem and a second-order recurrence.
- Extended the analysis to spread-out (bounded-step) and finite-variance walks, handling boundary overshoot to establish the $\asymp (x + 1)/r$ scaling for the ruin probability.

[†]Deferred due to mandatory military service.

[‡]Graduate-level course taken as an undergraduate.

[§]Supervised by Prof. [Yuki CHINO](#).

Credit Card Default Detection Using Machine Learning

February 2025 – June 2025

Final Group Project, [Machine Learning](#), École polytechnique

- Engineered predictive pipelines in Python (`scikit-learn`) to address imbalanced datasets via optimizing the F1-score to capture rare default events.
- Evaluated the trade-off between interpretability and accuracy by comparing baseline logistic regression against ensemble methods like XGBoost, demonstrating that transparent models can achieve comparable performance.

WORK EXPERIENCE

Teaching Assistant

September 2025 – June 2026 & September 2023 – June 2024

Department of Applied Mathematics, National Yang Ming Chiao Tung University

AY 2025 [Probability Theory](#) and [Calculus A \(II\)](#) under Prof. [FANG Xiang](#).

AY 2023 [Calculus A \(I\)](#) and [Calculus A \(II\)](#) under Prof. Yuki CHINO.

Private Tutor

March 2026 – June 2026, September 2024 – June 2025, & April 2023 – June 2023

Tutored high school and university students in mathematics, linear algebra, probability, and statistics with customized handouts.

Academic Tutor

October 2023 – November 2023 & April 2024 – May 2024

Center of Teaching and Learning Development, National Yang Ming Chiao Tung University

Tutored overseas students in Calculus A and conducted weekly sessions with customized handouts.

HONORS

Mathematics Presidential Award

March 2026, September 2023, & February 2023

Department of Applied Mathematics, National Yang Ming Chiao Tung University

For my academic performance in applied mathematics in my undergraduate study.

College Student Research Scholarship

July 2025 – February 2026

National Science and Technology Council

For the project “Study of Hitting Time and Probability for Random Walks in One and Two Dimensions” under research grant 114-2813-C-A49-139-M.

[PTSGI.com 23rd Translation Scholarship](#)

January 2026

[PTSGI.com](#) (Presidential Translation Service Group International)

Selected as one of the 30 recipients in the translation competition for Japanese-to-English translation.

College of Science Dean Award

October 2025

College of Science, National Yang Ming Chiao Tung University

For ranking in the top 10% of the department in my first three years of undergraduate study.

Outstanding Student Overseas Exchange Scholarship

June 2024

National Yang Ming Chiao Tung University & the Ministry of Education

For my exchange program at École polytechnique in France.

Honor Student

May 2023

National Yang Ming Chiao Tung University

For ranking in the top 5% of the department in my first semester of undergraduate study.